

A MINOR PROJECT PROPOSAL REPORT ON

Derma Vision

AI-Powered Skin Disease Detection and Personalized Skincare Assistant

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Abstract

Skin diseases affect an estimated 1.9 billion people globally, yet access to qualified dermatological care remains severely limited in rural and economically disadvantaged regions. This proposal presents **DermaVision**, an AI-powered platform that integrates deep learning-based skin disease detection, automated skin type classification, allergy and ingredient sensitivity analysis, and long-term progress tracking into a single, secure web application. At the core of the system is an **EfficientNet-B0** convolutional neural network trained via transfer learning on the HAM10000 and ISIC datasets. To address the interpretability challenge in medical AI, **Grad-CAM** heatmaps are generated alongside each prediction. A rule-based ingredient analysis engine further identifies potential allergens in user-provided skincare product formulations. The platform prioritises data privacy through JWT-based authentication, password hashing, and AES-256 encryption of stored medical images. Built with Streamlit on the frontend and deployed via Docker on cloud infrastructure, DermaVision aims to deliver an accessible, cost-effective, and clinically transparent dermatology support tool that reduces dependency on expensive specialist consultations for preliminary assessments.

Keywords: Artificial Intelligence, Deep Learning, EfficientNet-B0, Skin Disease Detection, Grad-CAM, Explainable AI, Personalized Skincare, Healthcare Security, Digital Dermatology, Computer Vision.

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1. Introduction

Skin diseases constitute one of the most prevalent categories of global health conditions, affecting individuals across all age groups, genders, and socioeconomic backgrounds. Disorders such as acne vulgaris, atopic dermatitis, psoriasis, superficial fungal infections, and contact dermatitis not only cause physical discomfort but also inflict measurable psychological harm, contributing to diminished self-esteem, anxiety, and reduced quality of life. The World Health Organization estimates that approximately 1.9 billion individuals worldwide suffer from skin conditions at any given time, yet fewer than 30% of low- and middle-income countries have an adequate density of practising dermatologists to address this burden.

The gap between disease prevalence and specialist availability is especially acute in Nepal and across the broader South Asian region, where geographical barriers, understaffed referral hospitals, and the prohibitive cost of consultation force many patients toward unverified online resources or untrained local practitioners. Delayed or incorrect diagnosis frequently allows treatable early-stage conditions to progress into chronic or complicated presentations, increasing long-term healthcare costs and patient suffering.

Recent advances in Artificial Intelligence — particularly Convolutional Neural Networks (CNNs) and Transfer Learning — have demonstrated performance approaching or matching experienced clinicians in dermatological image classification tasks [1]. However, a significant barrier to clinical adoption remains: most AI-based diagnostic tools function as opaque “black boxes”, offering no explanation of the reasoning behind their outputs. Explainable AI (XAI) techniques, and Grad-CAM in particular, address this shortcoming by generating visual attention heatmaps that allow both users and clinicians to interrogate and validate model predictions [5].

This proposal introduces **DermaVision** — a comprehensive, secure, and interpretable digital dermatology assistant. The system integrates AI-driven disease detection, skin type classification, ingredient-based allergy analysis, personalised skincare recommendations, and longitudinal skin health monitoring within a single web-based platform. By combining state-of-the-art deep learning with robust cybersecurity architecture and a user-centred design, DermaVision aims to democratise access to preliminary dermatological guidance and reduce the diagnostic burden on overstretched healthcare systems.

2. Motivation / Problem Statement

The following interconnected challenges motivate the development of DermaVision.

2.1 Limited Access to Qualified Dermatologists

Nepal has approximately 0.3 dermatologists per 100,000 people — a ratio far below the WHO-recommended threshold. In hill and mountain districts, specialist services are virtually absent. Patients who require early evaluation of suspicious skin lesions must undertake long-distance travel to Kathmandu or Pokhara, incurring costs and delays that routinely result in missed early-stage diagnoses.

2.2 Financial Barriers to Dermatological Consultation

Private dermatology consultations in Nepal range from NPR 1,500 to NPR 5,000 per visit, excluding diagnostic tests and prescription costs. For households below the national poverty line — representing over 20% of the population — this expense is prohibitive for conditions perceived as cosmetic rather than life-threatening.

2.3 Misleading and Unsupervised Self-Diagnosis

In the absence of accessible professional advice, individuals increasingly resort to search engines and social media for self-diagnosis. Studies have shown that online dermatological self-assessment results in incorrect conclusions in over 60% of cases, and frequently leads to inappropriate application of over-the-counter products that can aggravate the underlying condition.

2.4 Lack of Personalised Skincare Guidance

The global skincare industry is saturated with products marketed as universally effective, yet skin response to active ingredients varies enormously based on skin type, underlying conditions, and environmental exposure. Existing recommendation platforms overwhelmingly rely on demographic heuristics rather than individual physiological assessment.

2.5 Undetected Ingredient Sensitivities and Privacy Risks

Allergic contact dermatitis caused by cosmetic preservatives, fragrances, and surfactants is a growing clinical challenge, yet most consumers lack the biochemical literacy to identify potentially reactive ingredients. Concurrently, many consumer-facing skincare applications request photo uploads without implementing adequate encryption or access control, leaving sensitive dermatological images vulnerable to data breaches.

2.6 Absence of Integrated Longitudinal Monitoring

Dermatological treatment outcomes evolve over weeks or months, yet no widely available platform provides a unified interface combining initial diagnosis, treatment recommendation, and structured follow-up tracking.

3. Project Objective, Scope and Importance

3.1 Objectives

DermaVision is designed to achieve the following primary objectives:

1. Improve accessibility to preliminary dermatological assessment by delivering a web-based tool usable on any internet-connected device.
2. Achieve clinically useful diagnostic accuracy for common skin disease classification by fine-tuning EfficientNet-B0, targeting a minimum F1-score of 85% across all disease classes.
3. Build trust through explainability by integrating Grad-CAM visualisation so that patients and clinicians can inspect the spatial basis of each AI prediction.
4. Deliver personalised, condition-specific skincare recommendations derived from AI-classified skin type and detected condition.
5. Implement OWASP-compliant security architecture ensuring that sensitive dermatological data is protected through encryption, authentication, and access control.
6. Provide a longitudinal monitoring dashboard that enables users to track skin condition evolution across multiple assessment sessions over time.

3.2 Scope

1. AI-based classification of common skin diseases including acne, eczema, psoriasis, fungal infections, and selected pigmented lesions from the HAM10000/ISIC datasets.
2. Skin type classification (oily, dry, combination, sensitive, normal) from uploaded images and user-reported contextual information.
3. Ingredient and allergen screening against a curated database of known irritants.
4. Personalised skincare routine generation based on classified skin type, detected condition, and allergy profile.
5. Grad-CAM heatmap generation for interpretability of disease detection predictions.
6. Secure user authentication, encrypted image storage, and HTTPS-only data transmission.
7. A longitudinal progress tracking dashboard for monitoring skin condition changes.

8. Web-based deployment via Docker on cloud infrastructure.

3.3 Importance of the Project

Clinically: AI-assisted preliminary skin assessment with Grad-CAM visual explanations, enabling informed decisions about seeking care and supporting early detection of conditions such as melanoma.

Socially: Removal of cost and geographic barriers, benefiting rural communities and underserved populations in Nepal where specialist dermatology services remain limited.

Academically: Original integration of deep learning, explainable AI, recommendation systems, and cybersecurity applied to a healthcare problem, advancing understanding of AI model generalisation across South Asian skin tones.

Institutionally: Demonstration of student-led work at Nepal College of Information Technology achieving research-grade technical depth and social relevance under Pokhara University.

4. Literature Review

4.1 Deep Learning in Dermatological Image Classification

The landmark study by Esteva et al. [1], published in *Nature*, established that a CNN trained on 129,450 clinical images could classify keratinocyte carcinoma and melanocytic lesions with sensitivity and specificity statistically equivalent to 21 board-certified dermatologists. Codella et al. [2] formalised this research through the ISIC challenge, providing standardised benchmark datasets and evaluation protocols. Tschandl et al. [3] contributed the HAM10000 dataset, a large collection of multi-source dermatoscopic images that serves as a primary training resource for DermaVision’s classification model.

4.2 EfficientNet and Model Efficiency

Tan and Le [4] introduced EfficientNet, a family of CNN architectures built around a compound scaling principle that simultaneously increases network depth, width, and input resolution by a fixed ratio. EfficientNet-B0 achieves ImageNet top-1 accuracy of 77.1% with only 5.3 million parameters — a fraction of the parameters required by ResNet-50 or InceptionV3 for comparable performance [7].

4.3 Explainable AI and Clinical Trust

Rudin [6] terms the inability of high-capacity neural networks to articulate the basis of their predictions the “black box problem”. Selvaraju et al. [5] proposed Grad-CAM as a solution: by computing the gradient of the class score with respect to the final convolutional feature maps, the technique produces a coarse localisation heatmap highlighting regions most responsible for a given prediction.

4.4 Data Privacy and Security in Digital Health

The OWASP Foundation [9] identifies insufficient authentication mechanisms and unencrypted data storage as the leading causes of healthcare data breaches. The implementation of JWT-based session management, bcrypt-hardened password storage, and AES-256 encryption in DermaVision directly addresses these documented vulnerabilities [8].

4.5 Research Gap

Published dermatological AI tools treat security and longitudinal monitoring as peripheral concerns. None of the leading open-source or commercial systems surveyed implement healthcare-grade encryption (AES-256), token-based authentication (JWT), and secure password storage (bcrypt) alongside continuous lesion progress tracking. DermaVision

directly addresses this gap.

5. Methodology / Technical Description

The project adopts an **incremental development model**. The high-risk AI component (the skin disease detection model) is built and validated independently before dependent components are constructed, and the broad feature set benefits from phased integration.

5.1 System Architecture

DermaVision follows a five-tier layered architecture designed to ensure modularity, separation of concerns, and independent scalability of each component. Each layer communicates only with its adjacent layers through well-defined interfaces, minimising coupling and maximising maintainability.

At the lowest level, the **Storage Layer** manages all persistent data within the system. Structured metadata — including user profiles, assessment records, prediction results, and recommendation logs — is stored in a PostgreSQL relational database. Medical images uploaded by users are encrypted using AES-256 before being written to the file system, ensuring that sensitive dermatological photographs remain protected even in the event of a storage-level breach. Encryption keys are derived from a server-side secret and are never transmitted to or stored on client devices.

The **Data Access Layer** abstracts all database operations behind an Object-Relational Mapping (ORM) interface implemented with SQLAlchemy. This layer provides connection pooling for efficient utilisation of database connections, parameterised query execution to prevent SQL injection vulnerabilities, and centralised transaction management that ensures data consistency across related operations. By isolating persistence logic from business rules, this layer enables the underlying database technology to be replaced with minimal impact on higher layers.

The **Application Logic Layer** contains the core business modules of the platform. This includes the EfficientNet-B0 convolutional neural network for skin disease classification, a lightweight CNN for skin type determination, the Grad-CAM heatmap generation engine for explainable AI, the personalised skincare recommendation engine, and the ingredient allergen analyser. Each module is implemented as an independent service with a well-defined API, allowing individual components to be updated, tested, or replaced without affecting the rest of the system.

The **API Gateway Layer** exposes the functionality of the application layer through RESTful endpoints built with FastAPI. Every incoming request is intercepted by the gateway, which performs JWT validation to verify the identity of the caller, enforces rate limiting to protect against abuse, and routes requests to the appropriate backend service.

This layer serves as the single entry point for all client communication, providing a uniform interface that decouples frontend consumers from backend implementation details.

The **Presentation Layer** provides the user-facing interface of the platform, built using the Streamlit framework. Users interact with the system through a responsive web dashboard that supports image upload, displays diagnostic results with superimposed Grad-CAM overlays, presents personalised skincare recommendations, and visualises assessment history through an interactive timeline. The presentation layer communicates exclusively with the API gateway and contains no direct access to data stores or application logic.

5.2 Technology Stack

Table 5.1: Technology Stack

Component	Technologies
AI / Deep Learning	PyTorch, EfficientNet-B0, Transfer Learning, Grad-CAM
Backend	Python, FastAPI / Flask, PostgreSQL
Frontend	Streamlit, Responsive Web UI
Security	JWT, bcrypt, AES-256, HTTPS
DevOps / Cloud	Docker, Azure / Render
Datasets	HAM10000, ISIC Archive, DermNet NZ

5.3 Development Increments

Increment 1 — Basic System Setup (Week 1): System requirements finalisation, architecture design, development environment, Streamlit UI shell, JWT-based user registration and login.

Increment 2 — Skin Disease Detection (Weeks 2–4): Dataset collection and preprocessing (resizing to 224×224 , normalisation, augmentation), EfficientNet-B0 transfer learning with Cross-Entropy Loss and Adam optimiser, model evaluation, REST API integration.

Increment 3 — Skin Type and Allergy Analysis (Week 5): Lightweight CNN for skin type classification, allergen database compilation, ingredient analysis module.

Increment 4 — Explainable AI and Recommendation (Week 6): Grad-CAM implementation for heatmap visualisation, content-based recommendation engine for personalised skincare routines.

Increment 5 — Security and Progress Tracking (Weeks 7–8): AES-256 encryption, input validation, HTTPS enforcement, longitudinal progress tracking dashboard.

5.4 Entity-Relationship Diagram

Figure 5.1 presents the Entity-Relationship diagram for DermaVision using Chen-like notation. The diagram shows seven entities with their attributes, primary keys (underlined), foreign keys (prefixed FK:), and the cardinality of relationships between them. The schema is designed to support the full assessment lifecycle: user registration, image upload and encrypted storage, disease prediction with explainability metadata, personalised recommendation generation, and allergen tracking against a curated database.

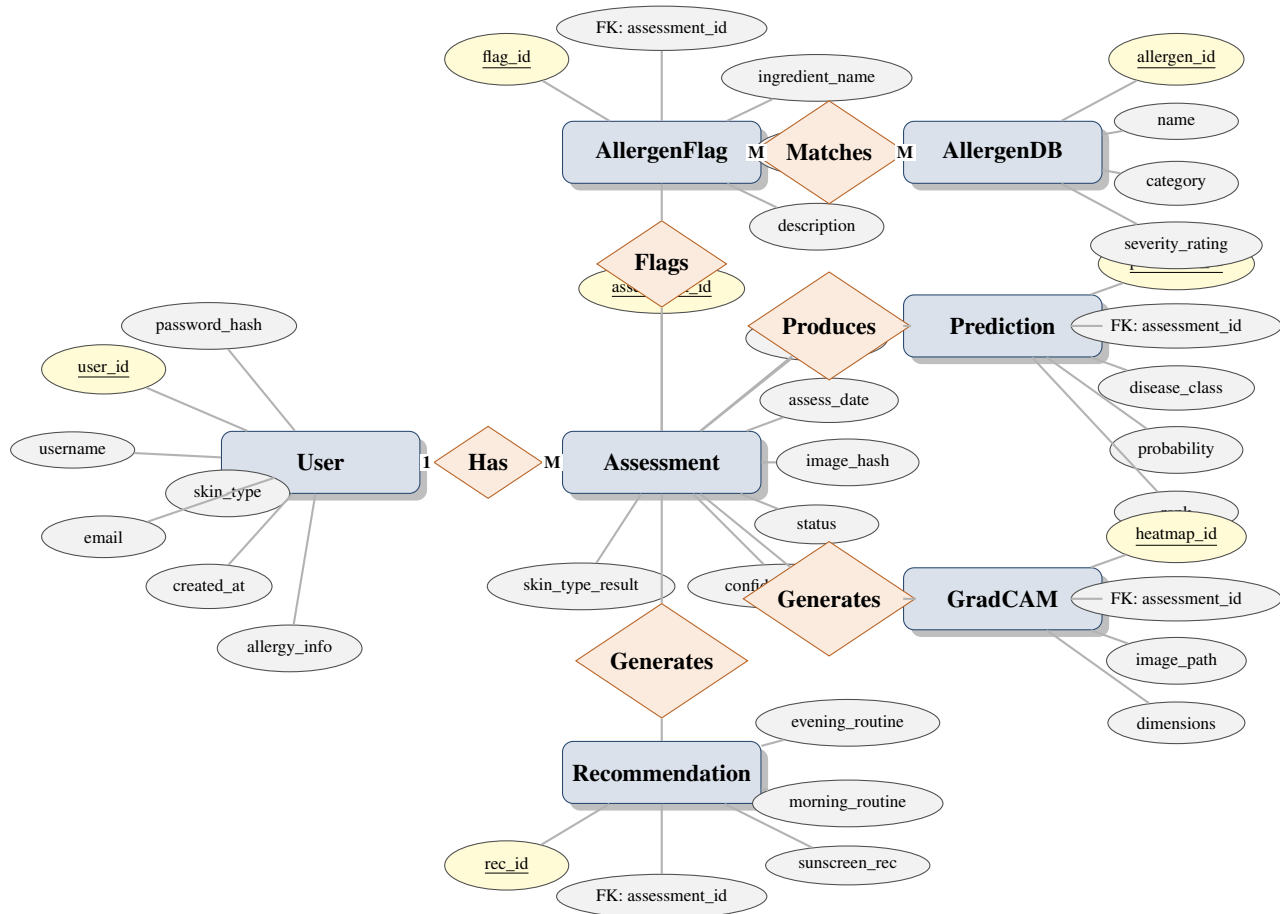


Figure 5.1: Entity-Relationship Diagram — Database Schema (Chen Notation)

5.5 Data Flow Diagram

Figure 5.2 presents a level-1 Data Flow Diagram for DermaVision using Gane-Sarson notation. External entities (End User, System Administrator) interact with the system through numbered processes. Data stores represent the persistent repositories that support each stage of the assessment lifecycle.

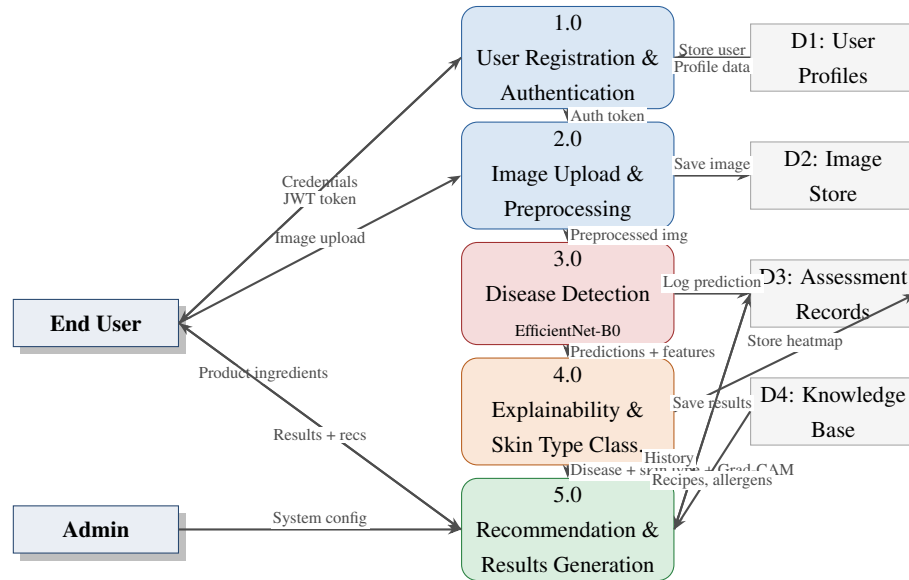


Figure 5.2: Level-1 Data Flow Diagram — Gane-Sarson Notation

5.6 Use Case Diagram

Figure 5.3 presents the Use Case diagram for DermaVision. The system defines two actors: the **End User** who interacts with the platform to perform skin assessments and track progress, and the **AI Detection Engine** which executes the core machine learning inference tasks. Use cases are grouped by functional area, and the relationships between them — including «include» and «extend» — capture dependencies among system behaviours.

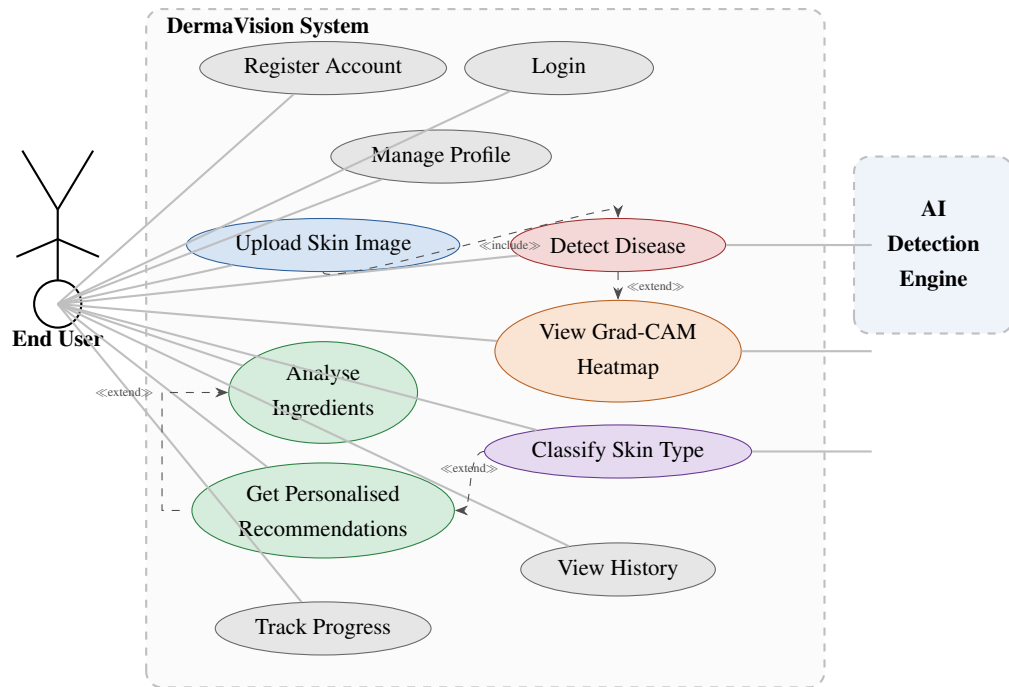


Figure 5.3: Use Case Diagram — System Actors and Core Functionality

6. Deliverables / Expected Outcomes

Upon successful completion of the project, the following outcomes are expected:

1. A fully functional web application integrating skin disease detection, Grad-CAM explainability, skin type classification, allergy analysis, personalised recommendations, and progress tracking.
2. A trained EfficientNet-B0 model validated on held-out test data with a minimum macro-averaged F1-score of 85% across all skin disease classes.
3. A Docker deployment on a cloud platform (Azure or Render), demonstrating real-world scalability and availability.
4. Comprehensive project documentation including this proposal, a final report, and a presentation prepared for the panel defence.
5. A scalable software architecture serving as a foundation for future enhancements including mobile application development, tele-dermatology integration, and expansion to additional disease categories.

7. Task and Time Schedule

The project spans nine weeks from 06 June 2026 to 05 July 2026, divided into eight structured phases.

Table 7.1: Project Phases and Duration

Phase / Activity	Period	Days
Requirement Analysis & Planning	Week 1	7
Dataset Collection & Preprocessing	Week 2	10
Model Development & Training	Weeks 3–4	13
Explainable AI Integration	Week 5	5
Backend & Security Implementation	Week 6	8
Frontend Development & Integration	Week 7	6
Feature Integration & Testing	Week 8	7
Deployment & Final Documentation	Week 9	7

7.1 Gantt Chart

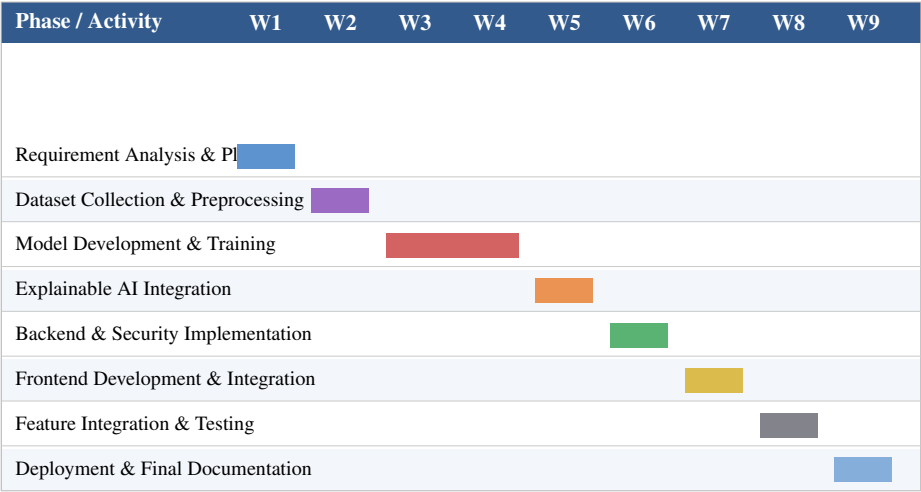


Figure 7.1: Gantt Chart — Timeline (W = Week)

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